

**Department of Computer Engineering**  
**Academic Term: July-November 2022**

**Class : B.E. Computer**  
**Semester : VII**

**Subject Name :ML**  
**Subject Code : CSC701**

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>Practical No:</b>        | 1                                        |
| <b>Title.</b>               | To Study and implement Linear Regression |
| <b>Date of Performance.</b> | 18-07-2025                               |
| <b>Roll No:</b>             | 9913                                     |
| <b>Name of the Student:</b> | Mark Lopes                               |

**Evaluation**

| <b>Indicator</b>           | <b>Very Poor</b>                  | <b>Poor</b>                       | <b>Average</b>        | <b>Good</b>                          | <b>Excellent</b>          |
|----------------------------|-----------------------------------|-----------------------------------|-----------------------|--------------------------------------|---------------------------|
| <b>Timeline (2)</b>        | More than three sessions late (0) | More than two sessions late (0.5) | Two sessions late (1) | One session late (1.5)               | Early or ontime (2)       |
| <b>Efforts(4)</b>          | N/A                               | N/A                               | Not Completed (1)     | Partially Completed (2)              | Completed(4)              |
| <b>Legibility(2)</b>       | N/A                               | N/A                               | Poor(1)               | Good(1.5)                            | Very Good(2)              |
| <b>Oral Assessment (2)</b> | N/A                               | N/A                               | N/A                   | Partially Understood the concept (1) | Understood the concept(2) |

```
In [ ]: import numpy as nm
import matplotlib.pyplot as mtp
import pandas as pd
```

```
In [ ]: file = pd.read_csv('/content/sample_data/Salary_Data.csv')
```

```
In [ ]: x = file[['YearsExperience']]
y = file['Salary']

print("X and Y : ")
print(x.shape)
print(y.shape)
```

X and Y :  
(30, 1)  
(30,)

```
In [ ]: from sklearn.model_selection import train_test_split
```

```
In [ ]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=0)

print("\nX_TRAIN and Y_TRAIN : ")
print(x_train.shape)
print(y_train.shape)

print("\nX_TEST and Y_TEST : ")
print(x_test.shape)
print(y_test.shape)
```

X\_TRAIN and Y\_TRAIN :  
(24, 1)  
(24,)

X\_TEST and Y\_TEST :  
(6, 1)  
(6,)

```
In [ ]: from sklearn.linear_model import LinearRegression
```

```
In [ ]: model = LinearRegression()

model.fit(x_train, y_train)

y_pred = model.predict(x_test)

print("\nModel predict y values: ")
print(y_pred)
```

Model predict y values:  
[ 40748.96184072 122699.62295594 64961.65717022 63099.14214487  
115249.56285456 107799.50275317]

```
In [ ]: mtp.scatter(x_train, y_train, color = 'red', label = 'Training Data')
mtp.plot(x_train, model.predict(x_train), color = 'blue', label = 'Regression Line')
mtp.title('Salary v/s Experience (Training Set)')
```

The scatter plot displays the relationship between years of experience and salary for a training set. The x-axis, labeled 'Years of Experience', ranges from 0 to 10. The y-axis, labeled 'Salary', ranges from 40,000 to 120,000. Red dots represent individual data points, and a solid blue line represents the linear regression fit. The data shows a clear upward trend, indicating that as experience increases, the salary also tends to increase.

| Years of Experience | Salary |
|---------------------|--------|
| 1.0                 | 39000  |
| 1.2                 | 46000  |
| 2.0                 | 43000  |
| 2.2                 | 40000  |
| 2.8                 | 56000  |
| 3.0                 | 60000  |
| 3.2                 | 54000  |
| 3.5                 | 64000  |
| 3.8                 | 56000  |
| 4.0                 | 56000  |
| 4.2                 | 56000  |
| 4.5                 | 61000  |
| 4.8                 | 68000  |
| 5.0                 | 66000  |
| 5.2                 | 83000  |
| 5.8                 | 81000  |
| 6.0                 | 94000  |
| 6.8                 | 92000  |
| 7.0                 | 98000  |
| 8.0                 | 101000 |
| 8.2                 | 114000 |
| 9.0                 | 105000 |
| 9.5                 | 112000 |
| 10.5                | 122000 |

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```
In [ ]: # import sklearn.metrics as r2_score
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)

print("R2 Score : ")
print(r2)

c = model.intercept_
m = model.coef_

print("\nModel intercept and coef")
print(m, c)
```

R2 Score :  
0.988169515729126

Model intercept and coef  
[9312.57512673] 26780.09915062818

```
In [ ]: accuracy = []

for i in range(100):
    x_train1, x_test1, y_train1, y_test1 = train_test_split(x, y, test_size = 0.8, ra
    model1 = LinearRegression()
    model1.fit(x_train1, y_train1)
    y_pred1 = model.predict(x_test1)
    r2_1 = r2_score(y_test1, y_pred1)
    accuracy.append(r2_1)
```

```
print(accuracy)
print(nm.max(accuracy))
```

```
[0.9536645755326943, 0.9587276669369498, 0.9599226033873139, 0.9474003096085318, 0.9
621063968368476, 0.9604183789115188, 0.9512638404910294, 0.9593236934450713, 0.95864
64746719688, 0.9412976253548547, 0.9596455139722875, 0.9645945039362755, 0.954645182
8306478, 0.9524283843195558, 0.9574349109881829, 0.9616217677370656, 0.9581808041466
211, 0.9583546185222949, 0.9641428142717868, 0.9616813005555789, 0.9443891385389038,
0.9578347285706635, 0.9509390773626878, 0.9627928948960339, 0.9524032048211271, 0.95
67653204443508, 0.9636873613970725, 0.9612603275880329, 0.9512660019542469, 0.962162
4564917671, 0.9526594452373836, 0.9557050174779026, 0.9638773784528749, 0.9497596777
951478, 0.9520178577454619, 0.9609223936735086, 0.9572491005122411, 0.95870004882875
36, 0.9636878446739479, 0.9422185509273002, 0.9501560722560173, 0.9510736292338142,
0.9639970519061158, 0.9662264814449496, 0.9503771843706749, 0.9473614276653719, 0.96
00741369543193, 0.9574501126663743, 0.96173672184421, 0.9600744775743649, 0.96146837
35473679, 0.950494768868645, 0.9579475923300533, 0.9570848179401535, 0.9526204223354
385, 0.9601767789630493, 0.9529857750528089, 0.938427688033362, 0.9566097445611071,
0.9695111211997747, 0.9579177248530238, 0.9529045551559473, 0.9650129321576381, 0.96
07690590975153, 0.9558401079078208, 0.953821050913366, 0.9633530805881929, 0.9525314
94217113, 0.9596164770706567, 0.9542990700558682, 0.9518009398927234, 0.962840894754
048, 0.9612347425144955, 0.9630980566538064, 0.939956073561345, 0.9663083377924078,
0.9506173252358386, 0.9522705290853881, 0.9551200629157734, 0.9534738892946083, 0.95
91085518943221, 0.9572519800510758, 0.9384132231966089, 0.9581215157168369, 0.958166
5113069906, 0.9595025129155997, 0.9549392840822342, 0.9525312590068257, 0.9539815539
857341, 0.965872795774837, 0.9494500804095631, 0.9470786930345976, 0.953711693254650
7, 0.9524522583709272, 0.9409849345516205, 0.9530882303436213, 0.9507159559904956, 0
.9563279411484026, 0.9515636676117039, 0.9534797597250289]
0.9695111211997747
```

```
In [ ]: ##### POSTLAB 1 #####
```

```
In [ ]: ### Q.1)

import numpy as np
import matplotlib.pyplot as mtp
import pandas as pd
import scipy.stats as sp
mtp.rcParams['figure.figsize'] = (20.0, 10.0)

df = {
    'X' : [10,20,30,40,50,60,70,80,90],
    'Y' : [420,365,285,220,176,117,69,34,5]
}

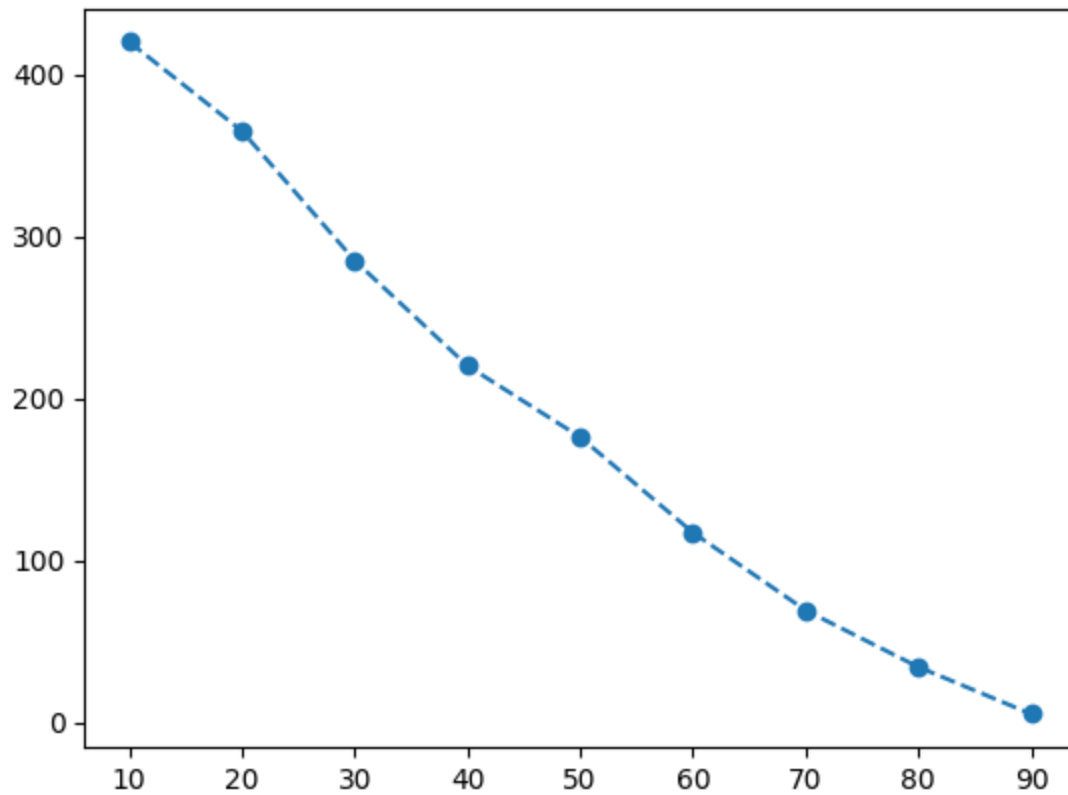
data = pd.DataFrame(df)
print(data.head(9))
```

|   | X  | Y   |
|---|----|-----|
| 0 | 10 | 420 |
| 1 | 20 | 365 |
| 2 | 30 | 285 |
| 3 | 40 | 220 |
| 4 | 50 | 176 |
| 5 | 60 | 117 |
| 6 | 70 | 69  |
| 7 | 80 | 34  |
| 8 | 90 | 5   |

```
In [ ]: X = data['X'].values
Y = data['Y'].values

mtp.plot(X,Y,marker='o', ls='--')
```

```
Out[ ]: [<matplotlib.lines.Line2D at 0x7d407c2dfdd0>]
```



```
In [ ]: y=np.array(data['Y'], dtype=float)
x=np.array(data['X'], dtype=float)
slope, intercept, r_value, p_value, std_err =sp.linregress(x,y)
xf = np.linspace(min(x),max(x),100)
yf = (slope*xf)+intercept
print("slope = ",slope,"\n","intercept = ",intercept,"\n",'r = ', r_value**2, '\n',

slope = -5.313333333333335
intercept = 453.55555555555554
r = 0.9840369938137091
p = 1.5050387692160386e-07
s = 0.2557818184006401
```

```
In [ ]: data["expected"] = data["X"] * slope + intercept

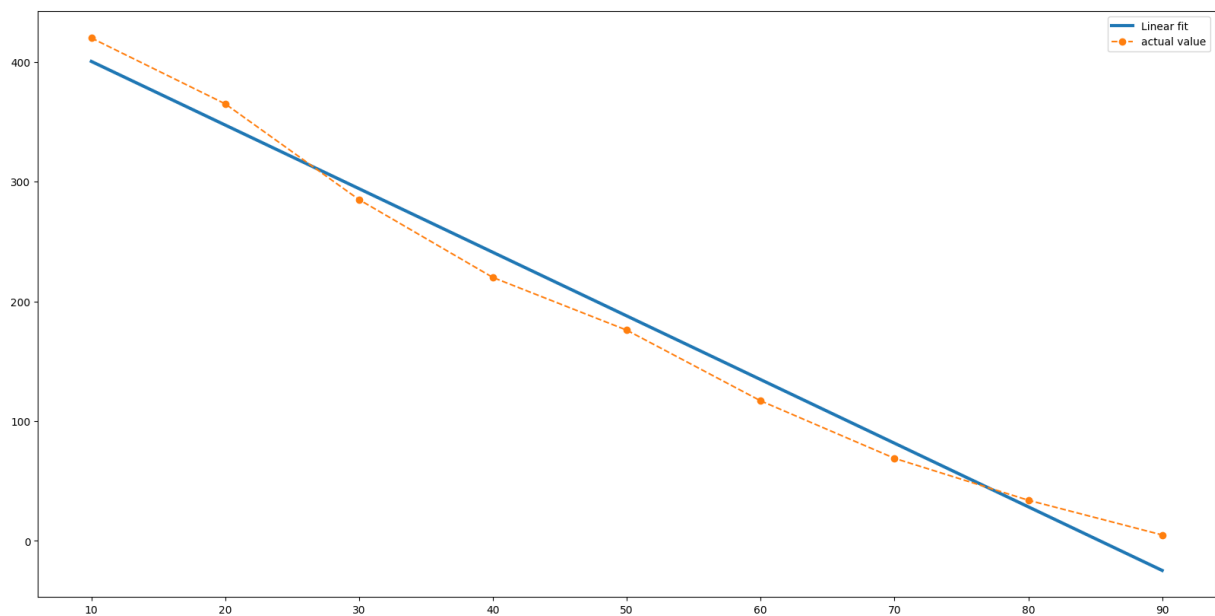
data.head(9)
```

```
Out[ ]:
```

|   | X  | Y   | expected   |
|---|----|-----|------------|
| 0 | 10 | 420 | 400.422222 |
| 1 | 20 | 365 | 347.288889 |
| 2 | 30 | 285 | 294.155556 |
| 3 | 40 | 220 | 241.022222 |
| 4 | 50 | 176 | 187.888889 |
| 5 | 60 | 117 | 134.755556 |
| 6 | 70 | 69  | 81.622222  |
| 7 | 80 | 34  | 28.488889  |
| 8 | 90 | 5   | -24.644444 |

```
In [ ]: f, ax = mtp.subplots(1, 1)
ax.plot(xf, yf, label='Linear fit', lw=3)
ax.plot(X, Y, label="actual value", marker='o', ls='--')
mtp.ylabel('')
ax.legend()
```

```
Out[ ]: <matplotlib.legend.Legend at 0x7d407ccf8650>
```



```
In [ ]: from scipy.stats import t
tinv = lambda p, df: abs(t.ppf(p/2, df))
ts = tinv(0.05, len(X)-2)
```

```
In [ ]: print(f"slope (95%): {slope:.6f} +/- {ts*std_err:.6f}")
```

```
slope (95%): -5.313333 +/- 0.604828
```

```
In [ ]: print(f"intercept (95%): {intercept:.6f}"f" +/- {ts*std_err:.6f}")
```



intercept (95%): 453.555556 +/- 0.604828

In [ ]: ##### Question 2 #####

```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import scipy.stats as sp
plt.rcParams['figure.figsize'] = (20.0, 10.0)

df = {
    'X':[0,25,50,75,100],
    'Y':[14,38,54,76,95]
}

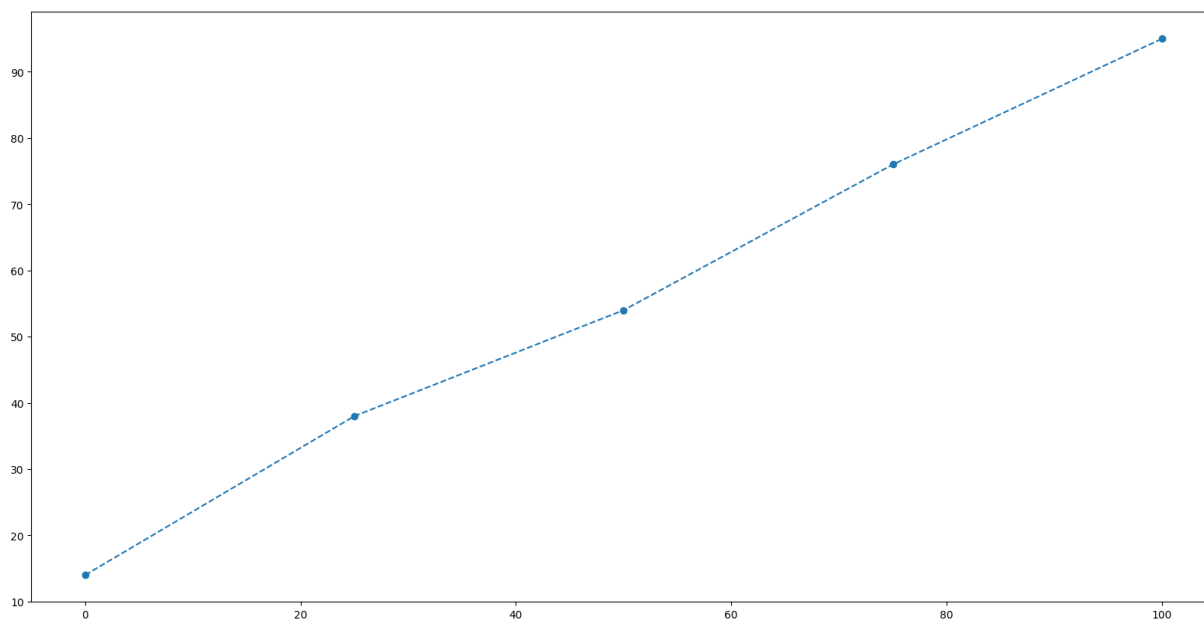
data = pd.DataFrame(df)
print(data.head(9))
```

|   | X   | Y  |
|---|-----|----|
| 0 | 0   | 14 |
| 1 | 25  | 38 |
| 2 | 50  | 54 |
| 3 | 75  | 76 |
| 4 | 100 | 95 |

```
In [ ]: X = data['X'].values
Y = data['Y'].values

plt.plot(X, Y, marker = 'o', ls = '--')
```

Out[ ]: [<matplotlib.lines.Line2D at 0x7d407beefdd0>]



```
In [ ]: y=np.array(data['Y'], dtype=float)
x=np.array(data['X'], dtype=float)
slope, intercept, r_value, p_value, std_err =sp.linregress(x,y)
xf = np.linspace(min(x),max(x),100)
yf = (slope*xf)+intercept
```

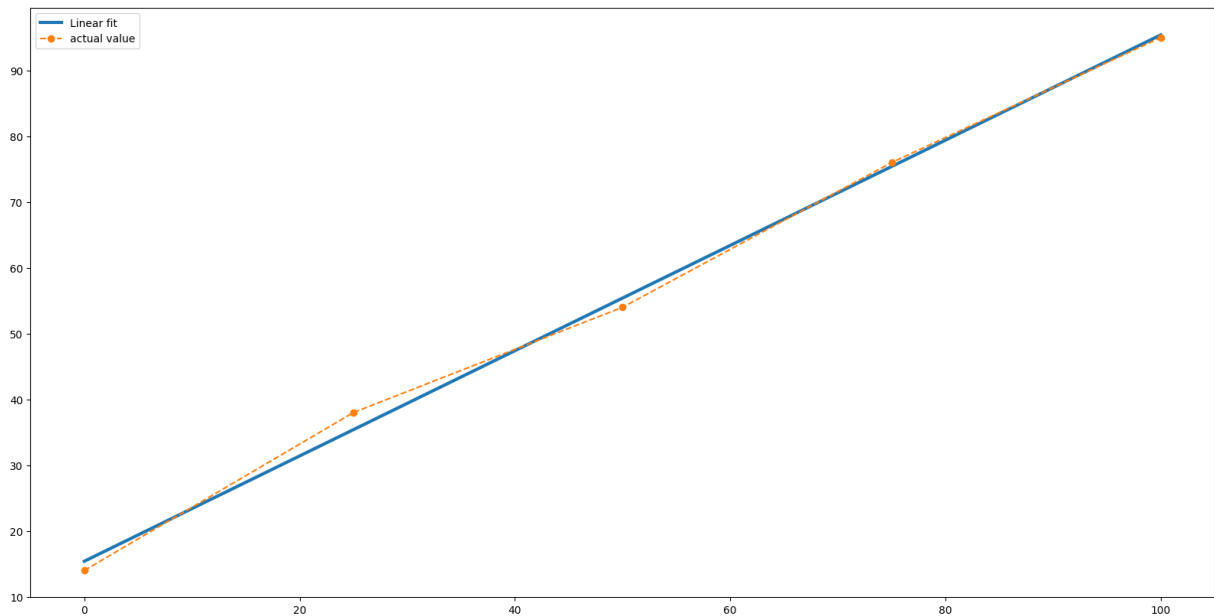
```
print("slope = ",slope,"\n","intercept = ",intercept,"\n",'r = ', r_value**2, '\n',
```

```
slope = 0.8
intercept = 15.399999999999999
r = 0.9972078181092939
p = 6.267128567572262e-05
s = 0.02444040370643135
```

```
In [ ]: data["expected"]=data["X"]*slope+intercept
print(data.head())
plt.rcParams['figure.figsize'] = (20.0, 10.0)
f, ax = plt.subplots(1, 1)
ax.plot(xf, yf,label='Linear fit', lw=3)
ax.plot(X,Y,label="actual value",marker='o', ls='--')
plt.ylabel('')
ax.legend()
```

|   | X   | Y  | expected |
|---|-----|----|----------|
| 0 | 0   | 14 | 15.4     |
| 1 | 25  | 38 | 35.4     |
| 2 | 50  | 54 | 55.4     |
| 3 | 75  | 76 | 75.4     |
| 4 | 100 | 95 | 95.4     |

```
Out[ ]: <matplotlib.legend.Legend at 0x7d407e7e0650>
```



```
In [ ]: from scipy.stats import t
tinvc = lambda p, df: abs(t.ppf(p/2, df))
ts = tinvc(0.05, len(X)-2)
print(f"slope (95%): {slope:.6f} +/- {ts*std_err:.6f}")
print(f"intercept (95%): {intercept:.6f} +/- {ts*std_err:.6f}")
```

```
slope (95%): 0.800000 +/- 0.077780
intercept (95%): 15.400000 +/- 0.077780
```

```
In [ ]: ##### Question 3 #####
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```

import scipy.stats as sp
plt.rcParams['figure.figsize'] = (20.0, 10.0)

df = {
    'X':[0,1,2,3,4],
    'Y':[2,3,5,4,6]
}

data = pd.DataFrame(df)
print (data.head(9))

X = data['X'].values
Y = data['Y'].values
plt.plot(X,Y,marker='o', ls='--')

y=np.array(data['Y'], dtype=float)
x=np.array(data['X'], dtype=float)
slope, intercept, r_value, p_value, std_err =sp.linregress(x,y)
xf = np.linspace(min(x),max(x),100)

yf = (slope*xf)+intercept
print("slope=",slope,"\n","intercept=",intercept,"\n",'r = ', r_value**2, '\n', 'p
print(f"\nequation of line: y={slope:.2f}x+{intercept:.2f}")

data["expected"]=data["X"]*slope+intercept
print(data.head())
plt.rcParams['figure.figsize'] = (20.0, 10.0)
f, ax = plt.subplots(1, 1)
ax.plot(xf, yf,label='Linear fit', lw=3)
ax.plot(X,Y,label="actual value",marker='o', ls='--')
plt.ylabel('')
ax.legend()

from scipy.stats import t
tinv = lambda p, df: abs(t.ppf(p/2, df))
pred = lambda x: slope*x+intercept
ts = tinv(0.05, len(X)-2)
print(f"slope (95%): {slope:.6f} +/- {ts*std_err:.6f}")
print(f"intercept (95%): {intercept:.6f}"f" +/- {ts*std_err:.6f}")
print("value of y at x=10=",pred(10))

```

```
      X  Y
0  0  2
1  1  3
2  2  5
3  3  4
4  4  6
slope= 0.9
intercept= 2.2
r = 0.81
p = 0.03738607346849863
s = 0.25166114784235827
```

equation of line:  $y=0.90x+2.20$

| X | Y | expected |
|---|---|----------|
|---|---|----------|

|   |   |   |     |
|---|---|---|-----|
| 0 | 0 | 2 | 2.2 |
|---|---|---|-----|

|   |   |   |     |
|---|---|---|-----|
| 1 | 1 | 3 | 3.1 |
|---|---|---|-----|

|   |   |   |     |
|---|---|---|-----|
| 2 | 2 | 5 | 4.0 |
|---|---|---|-----|

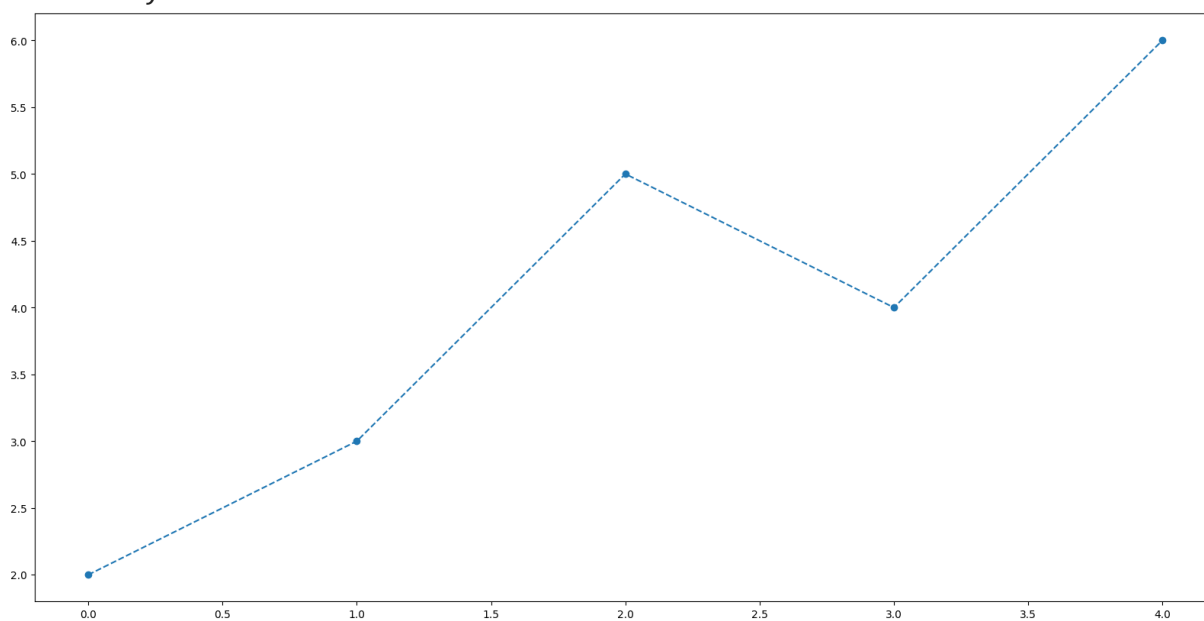
|   |   |   |     |
|---|---|---|-----|
| 3 | 3 | 4 | 4.9 |
|---|---|---|-----|

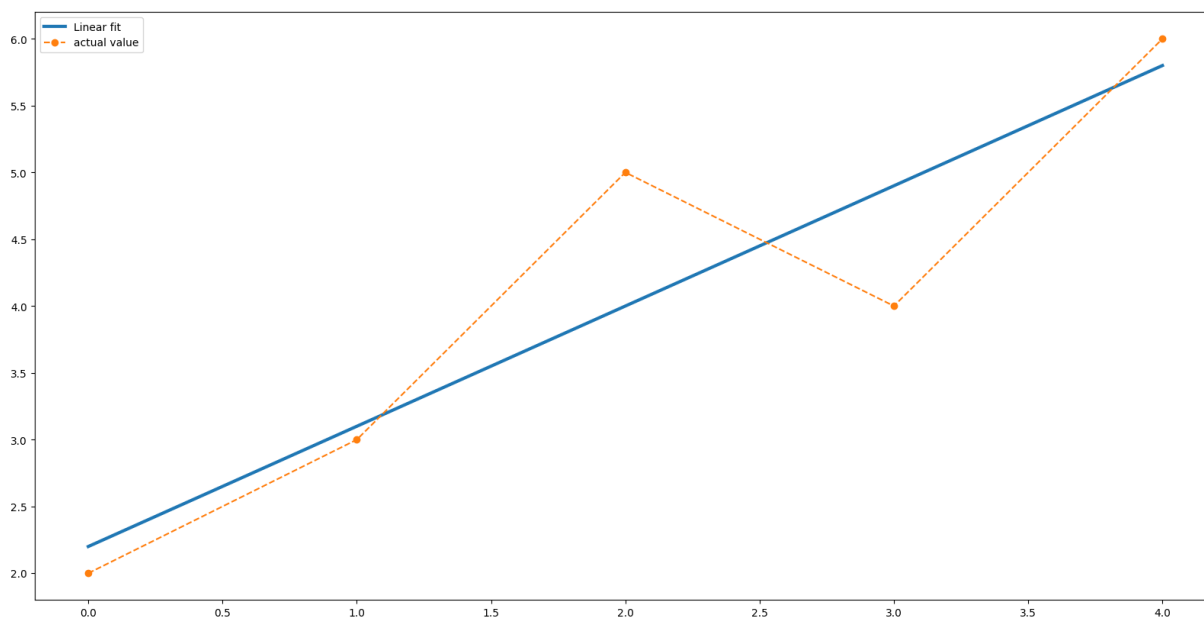
|   |   |   |     |
|---|---|---|-----|
| 4 | 4 | 6 | 5.8 |
|---|---|---|-----|

slope (95%):  $0.900000 \pm 0.800898$

intercept (95%):  $2.200000 \pm 0.800898$

value of y at x=10= 11.2





```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as sp
plt.rcParams['figure.figsize'] = (20.0, 10.0)

df = {'X':[2005,2006,2007,2008,2009],
      'Y':[12,19,29,37,45]}

data = pd.DataFrame(df)
print (data.head(9))

X = data['X'].values
Y = data['Y'].values
plt.plot(X,Y,marker='o', ls='--')

y=np.array(data['Y'], dtype=float)
x=np.array(data['X'], dtype=float)
slope, intercept, r_value, p_value, std_err =sp.linregress(x,y)
xf = np.linspace(min(x),max(x),100)

yf = (slope*xf)+intercept
print("slope=",slope,"\n","intercept=",intercept,"\n",'r = ', r_value**2, '\n', 'p
print(f"\nequation of line: y={slope:.2f}x+{intercept:.2f}")

data["expected"]=data["X"]*slope+intercept
print(data.head())
plt.rcParams['figure.figsize'] = (20.0, 10.0)
f, ax = plt.subplots(1, 1)
ax.plot(xf, yf,label='Linear fit', lw=3)
ax.plot(X,Y,label="actual value",marker='o', ls='--')
plt.ylabel('')
```

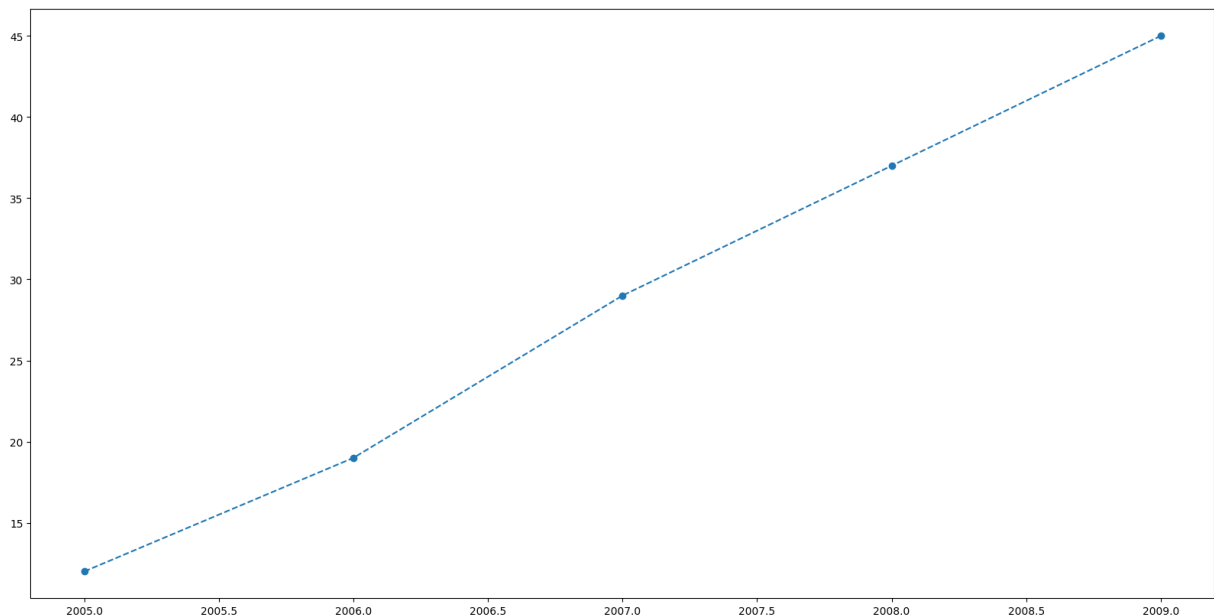
```
ax.legend()
```

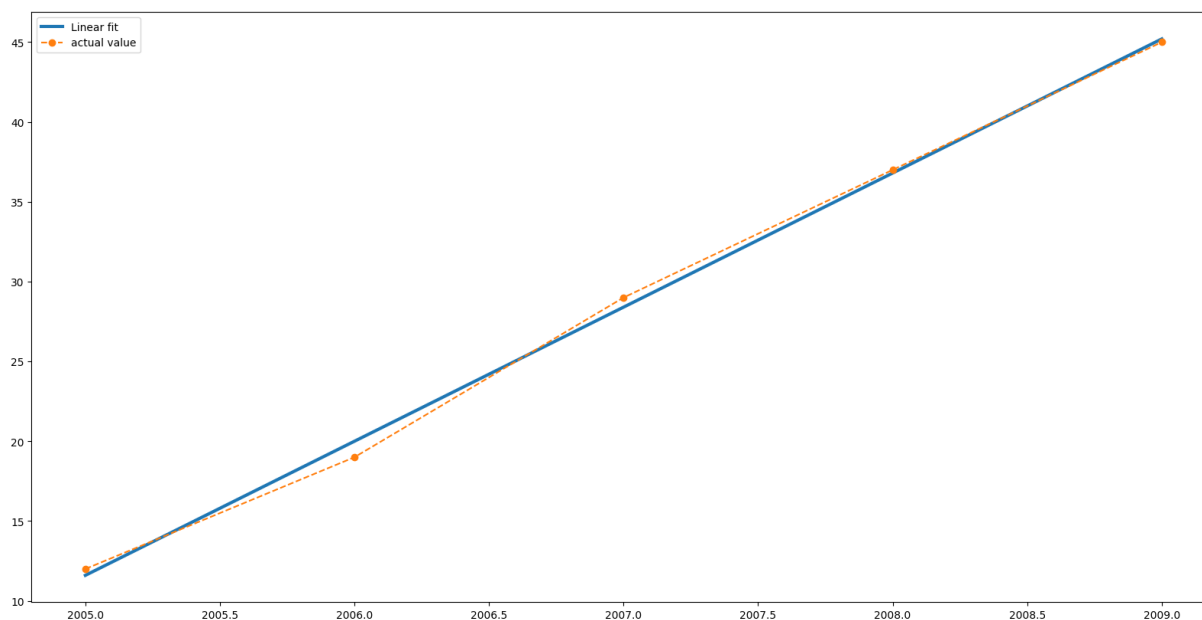
```
from scipy.stats import t
tinv = lambda p, df: abs(t.ppf(p/2, df))
pred = lambda x: slope*x+intercept
ts = tinv(0.05, len(X)-2)
print(f"slope (95%): {slope:.6f} +/- {ts*std_err:.6f}")
print(f"intercept (95%): {intercept:.6f}"f" +/- {ts*std_err:.6f}")
print("value of y at x=2012=",pred(2012))
```

```
      X    Y
0  2005   12
1  2006   19
2  2007   29
3  2008   37
4  2009   45
slope= 8.4
intercept= -16830.399999999998
r = 0.9977375565610856
p = 4.570360814080035e-05
s = 0.23094010767586692
```

equation of line:  $y=8.40x+-16830.40$

```
      X    Y  expected
0  2005   12     11.6
1  2006   19     20.0
2  2007   29     28.4
3  2008   37     36.8
4  2009   45     45.2
slope (95%): 8.400000 +/- 0.734954
intercept (95%): -16830.400000 +/- 0.734954
value of y at x=2012= 70.400000000000146
```





```
In [ ]: ##### Question 5 #####
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as sp
plt.rcParams['figure.figsize'] = (20.0, 10.0)

df = {'X':[368,340,665,954,331,556,376],
      'Y':[1.7,1.5,2.8,5,1.3,2.2,1.3]}

data = pd.DataFrame(df)
data=data.sort_values('X')
print (data.head(9))

X = data['X'].values
Y = data['Y'].values
plt.plot(X,Y,marker='o', ls='--')

y=np.array(data['Y'], dtype=float)
x=np.array(data['X'], dtype=float)
slope, intercept, r_value, p_value, std_err =sp.linregress(x,y)
xf = np.linspace(min(x),max(x),100)

yf = (slope*xf)+intercept
print("slope=",slope,"\n","intercept=",intercept,"\n",'r = ', r_value**2, '\n', 'p
print(f"\nequation of line: y={slope:.2f}x+{intercept:.2f}")

data["expected"]=data["X"]*slope+intercept
print(data.head())
plt.rcParams['figure.figsize'] = (20.0, 10.0)
f, ax = plt.subplots(1, 1)
ax.plot(xf, yf,label='Linear fit', lw=3)
```

```

ax.plot(X,Y,label="actual value",marker='o', ls='--')
plt.ylabel('')
ax.legend()

from scipy.stats import t
tinv = lambda p, df: abs(t.ppf(p/2, df))
pred = lambda x: slope*x+intercept
ts = tinv(0.05, len(X)-2)
print(f"slope (95%): {slope:.6f} +/- {ts*std_err:.6f}")
print(f"intercept (95%): {intercept:.6f}"f" +/- {ts*std_err:.6f}")
print("value of y at x=8=",pred(8))

```

|   | X   | Y   |
|---|-----|-----|
| 4 | 331 | 1.3 |
| 1 | 340 | 1.5 |
| 0 | 368 | 1.7 |
| 6 | 376 | 1.3 |
| 5 | 556 | 2.2 |
| 2 | 665 | 2.8 |
| 3 | 954 | 5.0 |

slope= 0.005606157184993036  
 intercept= -0.6180148991607144  
 r = 0.9612629035488399  
 p = 0.00010169537218360504  
 s = 0.0005032951082414453

equation of line:  $y=0.01x+-0.62$

|   | X   | Y   | expected |
|---|-----|-----|----------|
| 4 | 331 | 1.3 | 1.237623 |
| 1 | 340 | 1.5 | 1.288079 |
| 0 | 368 | 1.7 | 1.445051 |
| 6 | 376 | 1.3 | 1.489900 |
| 5 | 556 | 2.2 | 2.499008 |

slope (95%): 0.005606 +/- 0.001294  
 intercept (95%): -0.618015 +/- 0.001294  
 value of y at x=8= -0.5731656416807701

